

Sablefish age estimation

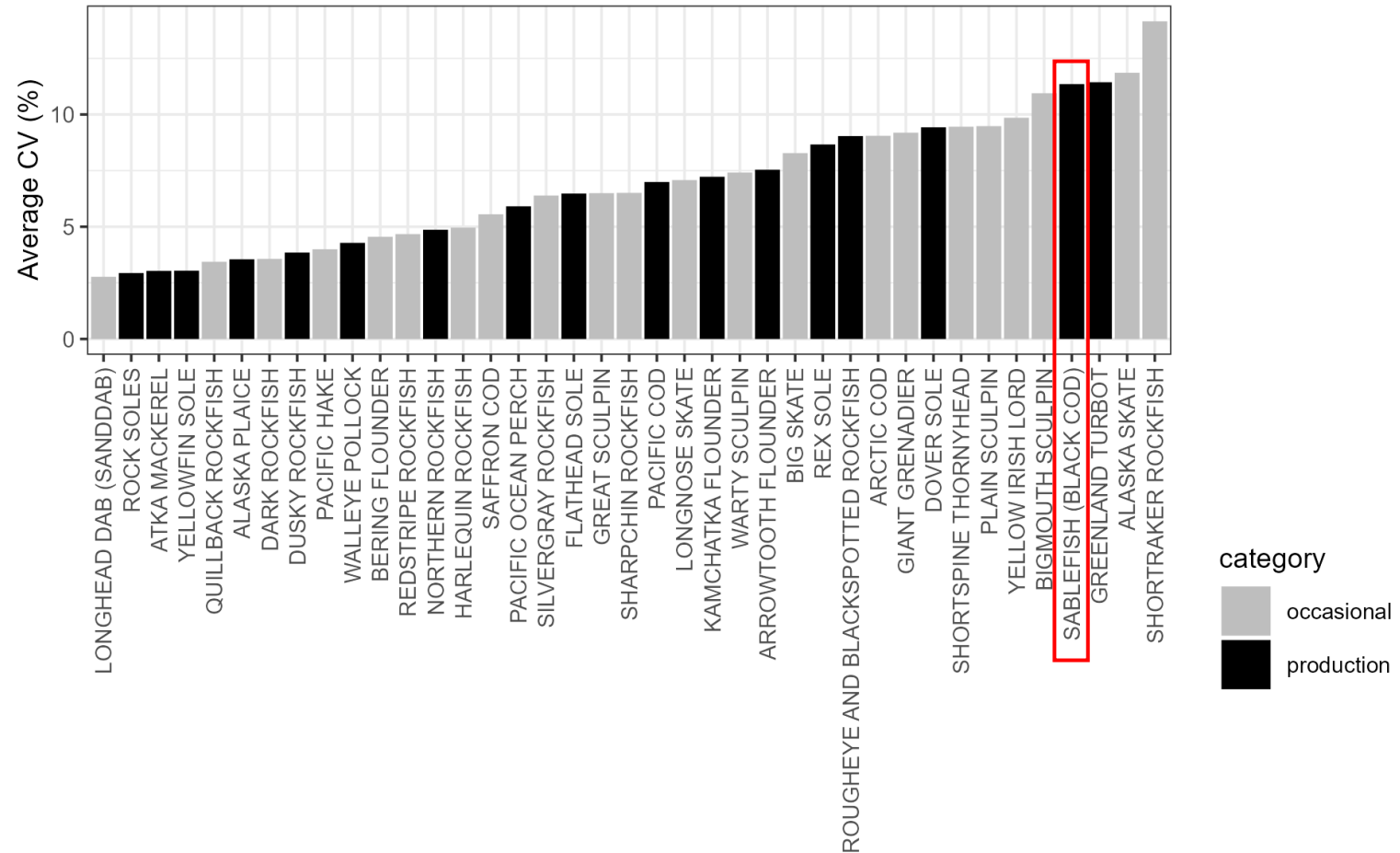
AFSC Age and Growth Program

ad hoc presentation to sablefish CIE review panel

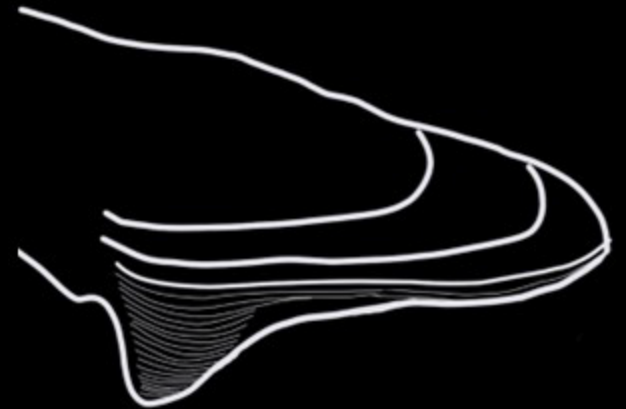
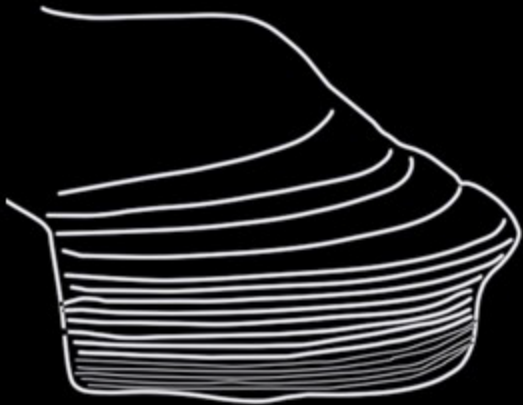
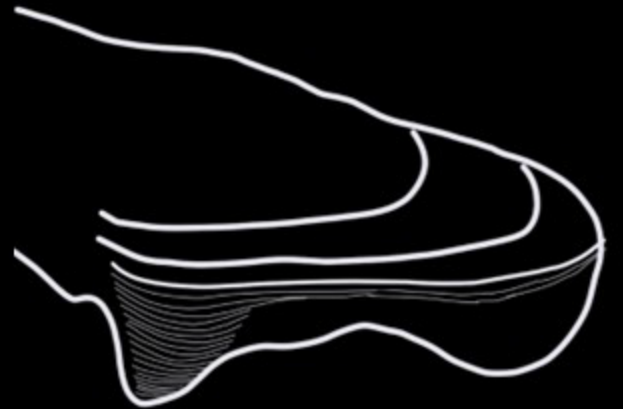
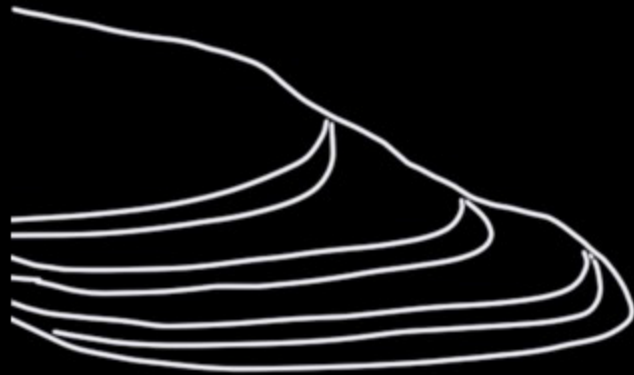
June 17, 2026

Sablefish otolith interpretation

- One of the most difficult species aged by the AFSC Age and Growth Program
- Checks difficult to discern from true annuli
- Sablefish otoliths have diverse shapes and patterns which makes interpretation difficult



Sablefish otolith shapes



Sablefish age validation studies

- OTC mark-recapture of adult fish
- Net-pen rearing
- Lead-radium disequilibria (radiometric dating)
- Tagged and recaptured known-age juveniles

Developing and applying age determination criteria

- Recovered otoliths from tagged juveniles (known-age otoliths) have helped us develop criteria for each otolith shape morph
- Member of Committee of Age Reading Experts (otolith exchanges)
- Training materials (written manual, in-house instructional video, and reference collection for maintaining consistency)

Agencies involved:

- Alaska Department of Fish and Game (ADFG)
- Washington Department of Fish and Wildlife (WDFW)
- Oregon Department of Fish and Wildlife (ODFW)
- Pacific States Marine Fisheries Commission (PSMFC)
- NOAA Fisheries Southwest Fisheries Science Center (SWFSC)



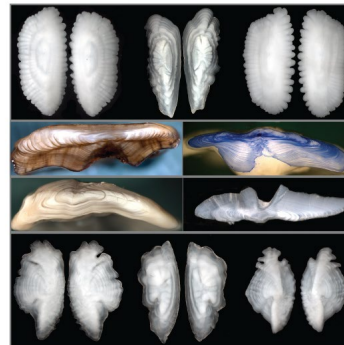
Agencies involved:

- International Pacific Halibut Commission (IPHC)
- Fisheries and Oceans Canada (DFO)
- California Department of Fish and Game (CDFG)
- NOAA Fisheries Alaska Fisheries Science Center (AFSC)
- NOAA Fisheries Northwest Fisheries Science Center (NWFSC)

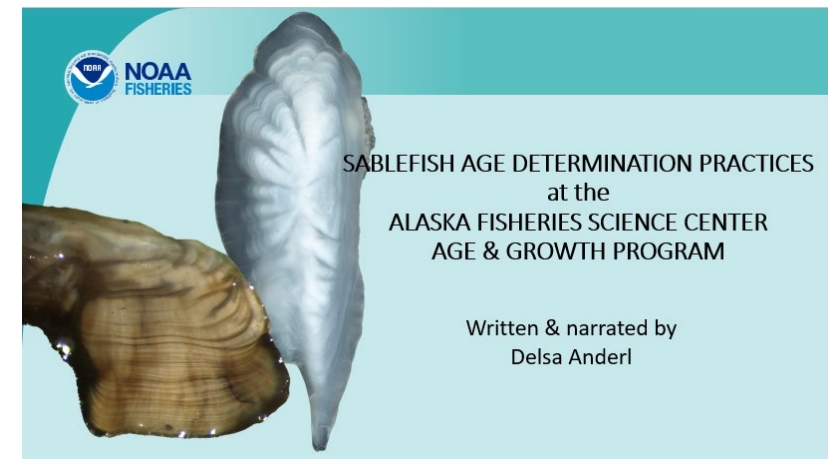
NOAA Professional Paper NMFS 13

Age determination manual of the Alaska Fisheries Science Center Age and Growth Program

Mary Elizabeth Matto and Daniel K. Kimura (editors)



U.S. Department of Commerce
February 2012



Examples of sablefish otolith growth patterns
from our known-age collection

Known Age II Sp # 270
8 yr old 700 mm
Date of recovery: Aug 7

Pattern of regular growth

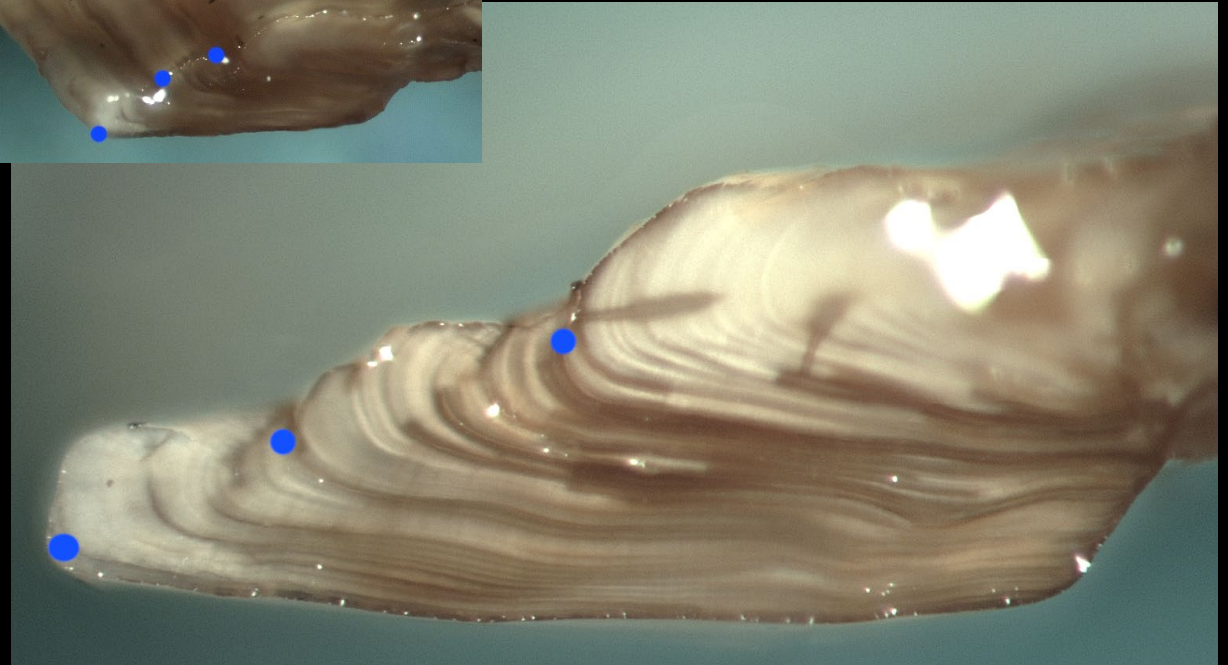
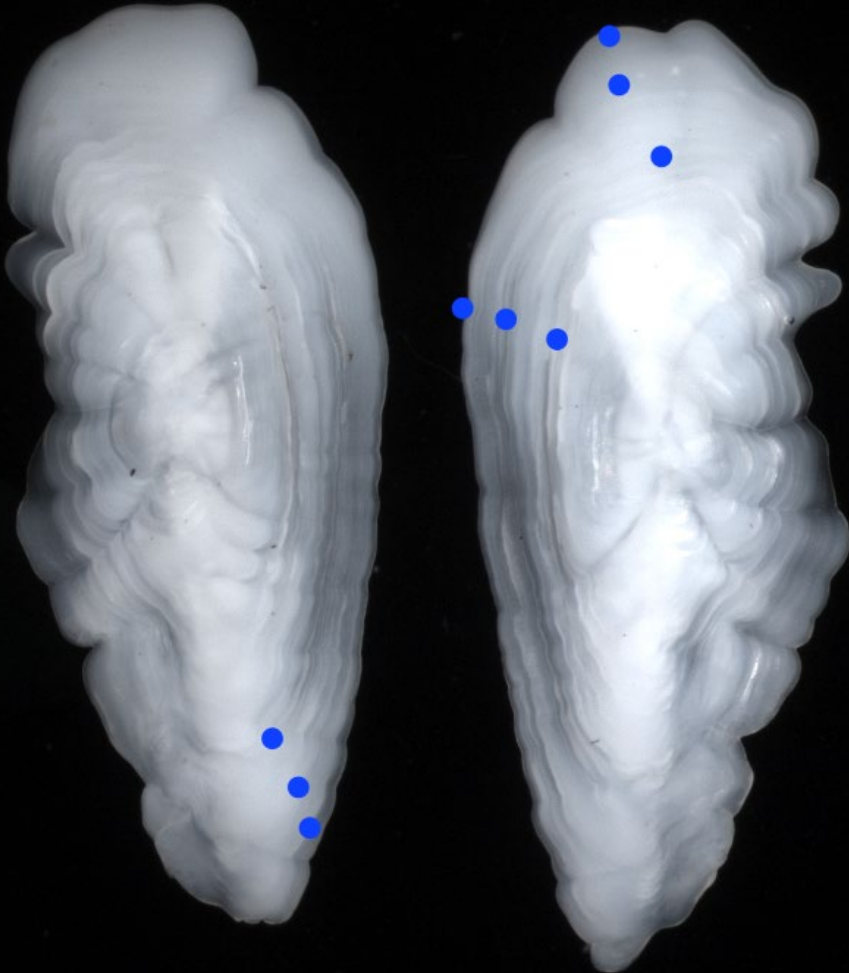


Known Age III Sp # 2923

3 yr old 560 mm

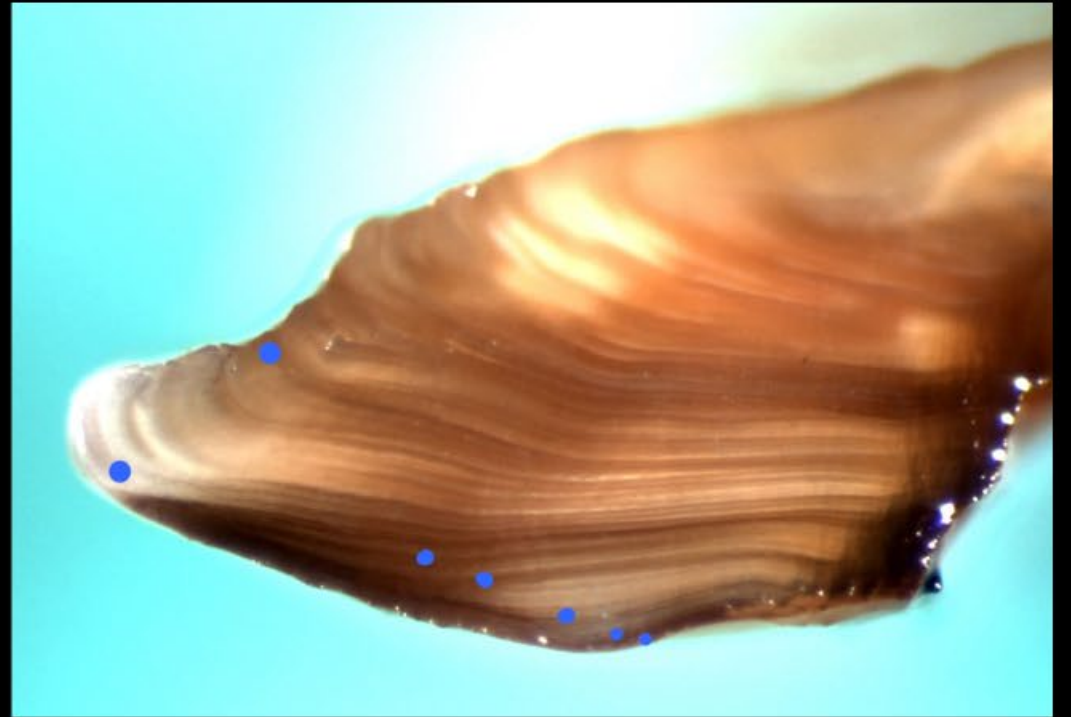
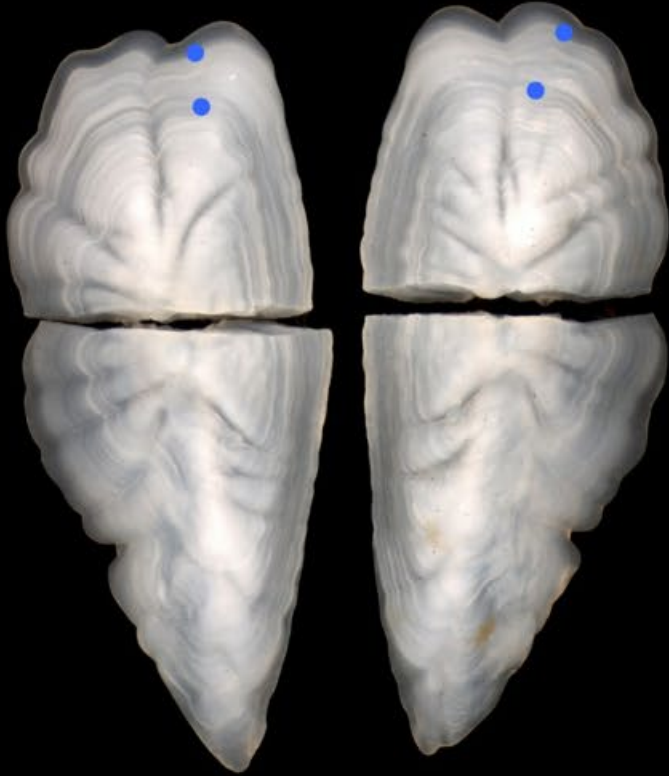
Date of recovery: Mar 30

Pattern of fast growth



Known Age II Sp # 70
7 yr old 500 mm
Date of recovery: Apr 27

Pattern from fast to slow growth

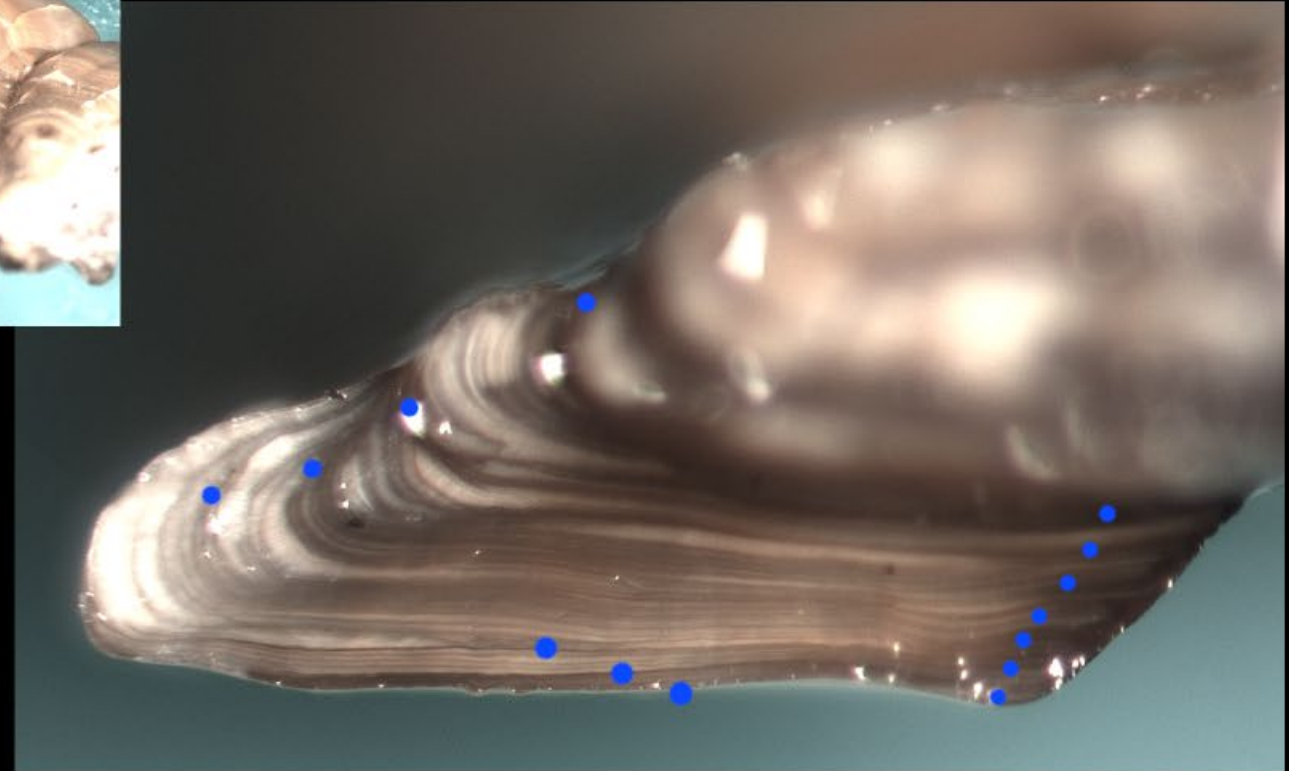
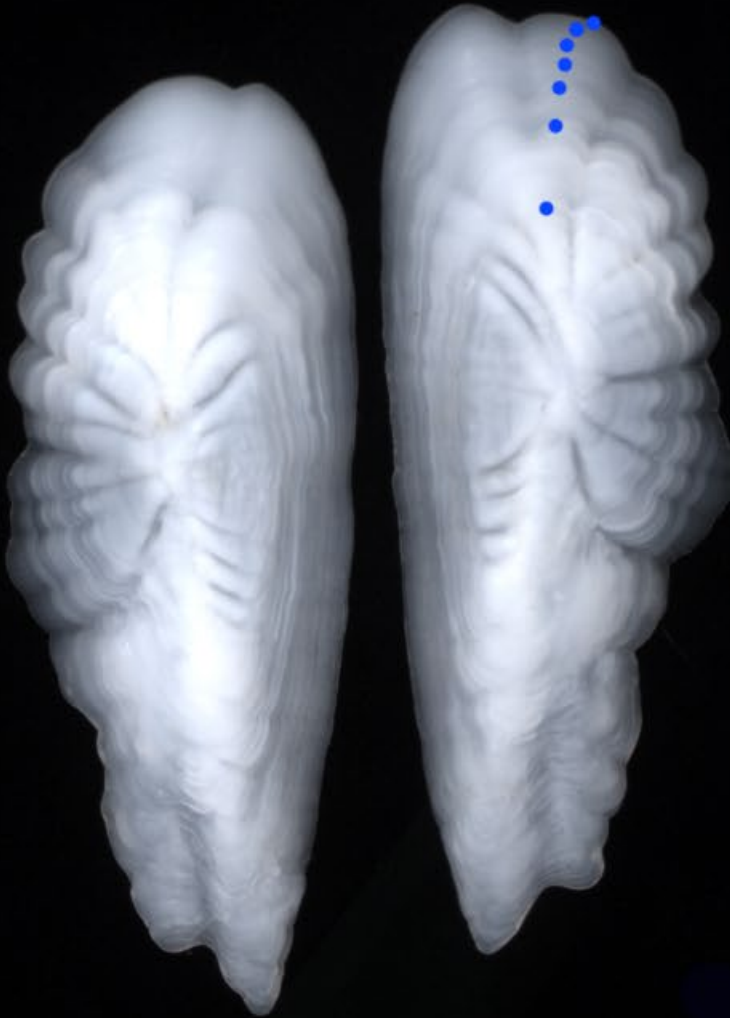


Known Age III Sp # 3221

7 yr old 610 mm

Date of recovery: Jun 10

Pattern of annuli over-growth



Pattern of fast growth transitioning to extreme slow growth

